

Tanay Chandak

Smart Medicine Box

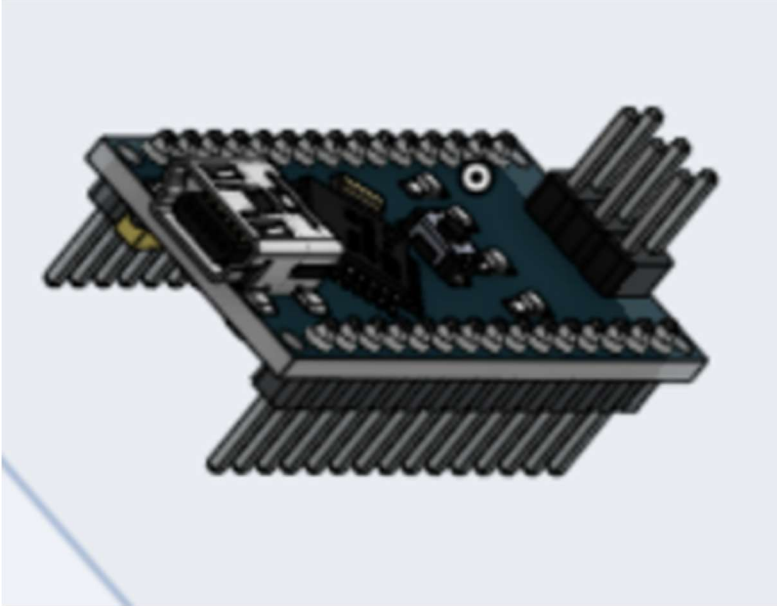
Introduction

I decided to create a medicine box with 10 compartments, a lid with a build in LED connected to an Arduino Nano that can take input from the user. The box is designed and proportioned to be easily handled by individuals with average anthropometric hand measurements and can hold multiple medicinal components and drugs based on individual needs. This smart medicine box is classifiable as a medical device since it aids in medication management on a timely manner (user inputs timings of drug intake and need) and conveys physical input to the user via the LED controlled by the Arduino system. Under the U.S. FDA and EU MDR, this device would classify as a Class 1 (low risk) Medical Device.

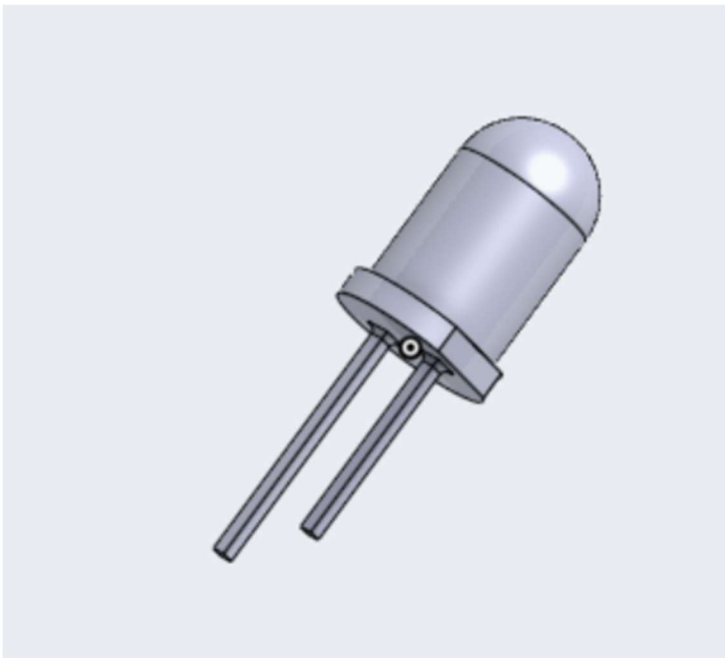
The central body of the medicine box is designed to be about 5 inches by 2 inches by 1.25 inches with a think lid that rotates about a hinge with an LED holder on top. In addition, the device contains a small compartment outside the main body to hold an Arduino Nano (dimensions of 48mm x 18mm x 22 mm) which I dimensioned to fit the Arduino with room for tolerance. Overall, the device is small so it is not very invasive and large enough such that it can hold adequate amounts of medicine required for an individual who will use it.

List of Parts:

- Main Compartment (contains many parts but not separated)
 - Contains the base of the box with the 10 compartment divisions.
 - Contains compartment for the Arduino Nano with hole for input
 - Contains 2 hinge bases for the lid rotation
 - Contains the lid for the Arduino component (separate part)
- Hinge Bar (for the rotation of the lid in the x and y axis while keeping it aligned in the z)
- Main Compartment Lid (contains many parts but not separated)
 - Contains the main lid for the 10 medicine compartments
 - Contains the compartment for the LED to sit in with a wiring hole
 - Contains the hinge base to join the lid and main compartment with mates
- Arduino Nano to actual scale

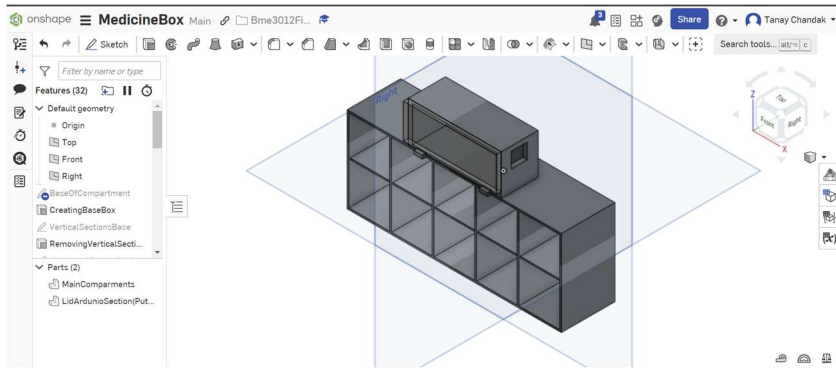


- Red LED with alterations to fit the device (Altered the led prong lengths to make them fit my design)



Part Images

Figure 1: Isometric View of the Main Compartment



In *Figure 1* you can see the main body of the medicine box. To limit the number of parts, I chose to combine simple parts into one such that the entire body (except 1 lid) moves as one part.

Figure 2: Isometric View of Hinge Rod

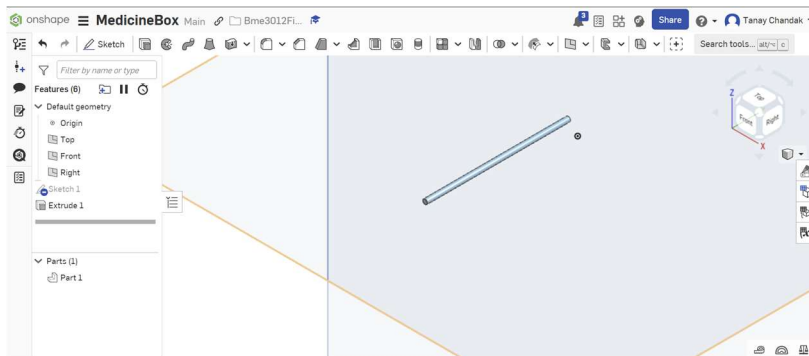
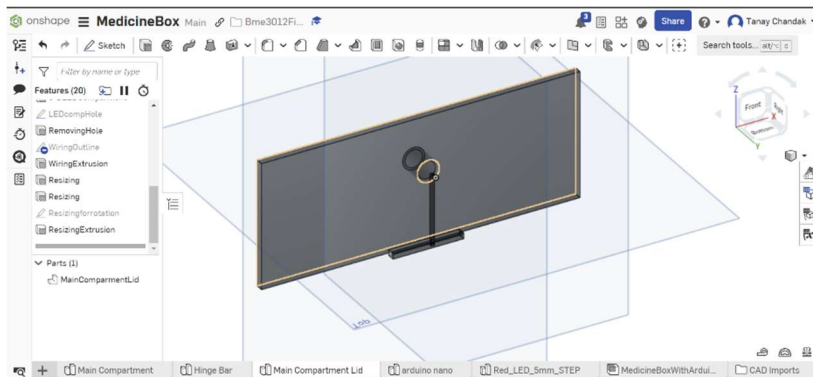


Figure 3: Isometric View of the Main Compartment Lid



In *Figure 3*, we can see the compartment lid and the individual components involved in it.

Assembly Pictures

Figure 4: Isometric View of the Device

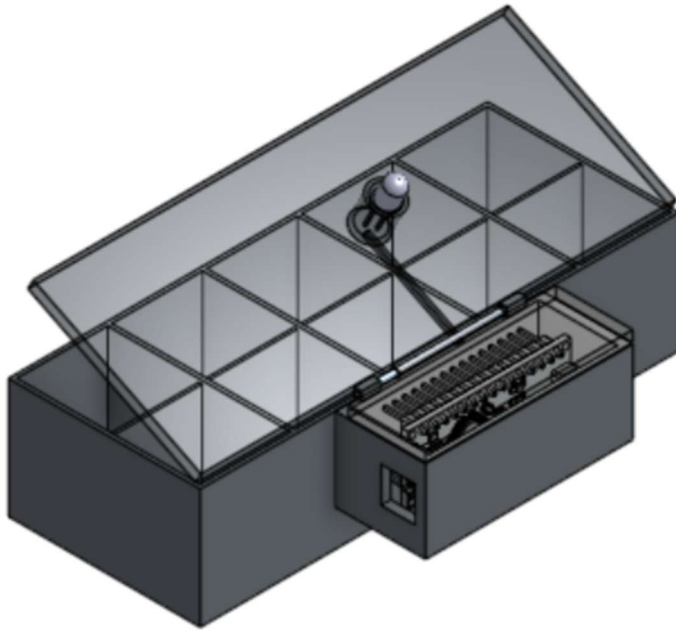


Figure 4 shows the completed design of the smart medical device. The LED and Arduino are not part of the design but have been incorporated into the design. The hinged transparent lid allows for clear visibility and easy access to the compartments, while maintaining protection against dust and contamination. The electronic housing is positioned on the exterior wall to safely enclose the circuitry and power source without interfering with the medication storage area. This design effectively combines mechanical functionality with electronic alert integration to create a compact, user-friendly medical adherence system.

Figures 5-7 provide further views of the device and Figure 8 provides an explanation of the mates used and their roles.

Figure 5: Front View

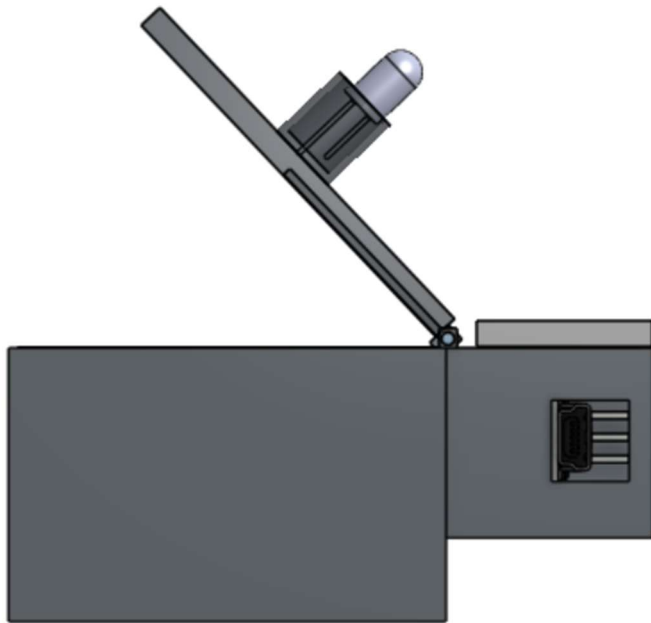


Figure 6: Top View

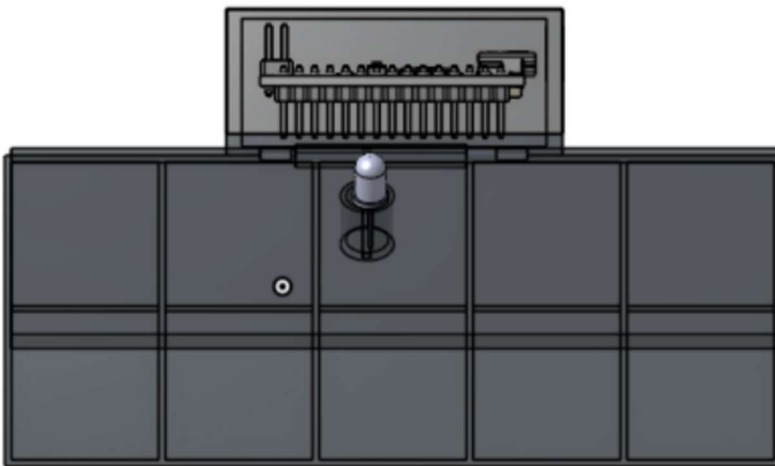


Figure 7: Closed Lid Isometric View

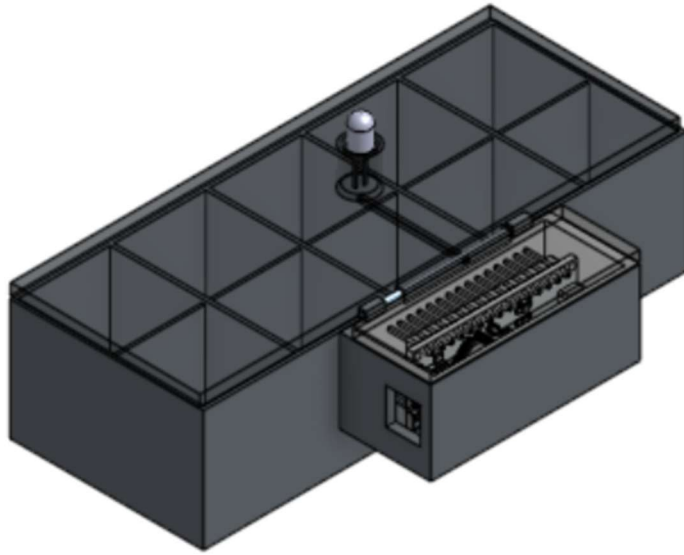
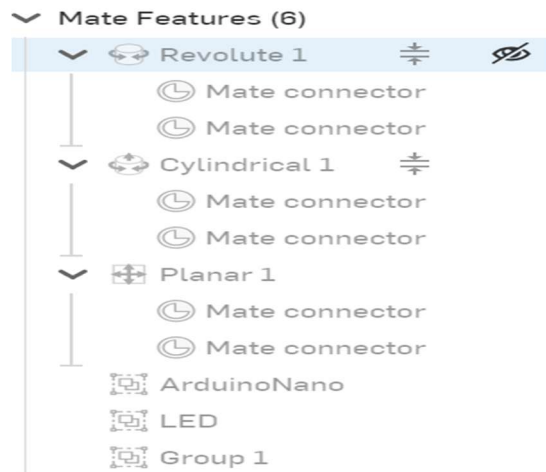


Figure 8: The Mates



1. Revolute Mate: This mate connects the hinge rod to the main compartment body allows for rotational motion around the fixed axis such that the hinge can only rotate about the z-axis. This mate enables the rod to act as a pivot point for the lid mechanism, replicating how an actual hinge would function in a physical prototype. It restricts all translational movement, permitting only rotation, which ensures realistic lid motion and proper hinge alignment.
2. Cylindrical Mate: This mate joins the lid to the hinge rod which has already been connected to the main body via the revolute mate. This mate enables both rotation and

limited sliding along the rod's axis. This allows for the hinge motion required for the lid to close and open but also allows for slight axial movement such to limit possible stress and friction that could occur if the parts rub. Realistically, this motion should be limited to a very miniscule distance so that the device still functions properly and the rotational motion should be limited between 0-150 degrees from the closed box (seen in figure 7) to a fully open box.

3. Planar Mate: The planar mate is used to align and fix the components of the box. This is to ensure the multiple parts share a reference plane and do not move off that plane. The purpose is to provide stability and prevent mis-alignments.

Together, these mates replicate the intended mechanical behavior of the medicine box: a smoothly hinged lid supported by a fixed rod and stable alignment between all major parts.

Figure 9: Medicine Container Primary Body Part Drawing

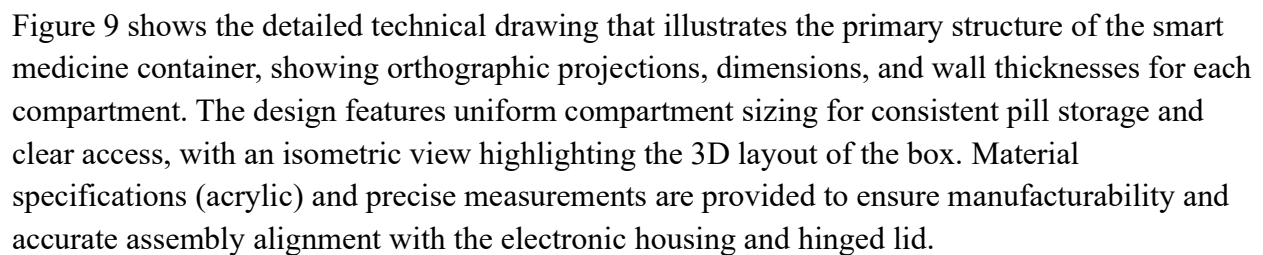


Figure 10: The Tolerance Levels and Other Constant Info (Applies to all Drawings)

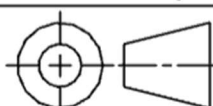
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .XX = ± 0.05 - .XXX = ± 0.005 - .XXXX = ± 0.0005 - SURFACE FINISH  ANGULAR = $\pm 1^\circ$ FRACTIONAL = $\pm 1/32$		NAME	DATE
	DRAWN	TANAY CHANDAK	10/30/2025
	CHECKED		
	APPROVED	---	---
DO NOT SCALE DRAWING			
BREAK ALL SHARP EDGES AND REMOVE BURRS			
THIRD ANGLE PROJECTION	MATERIAL		FINISH
	ACRYLIC		

Figure 10 simply shows the **tolerances** for the parts since it could be difficult to see on the larger drawings. Material and tolerances stay constant throughout the parts.

These specific tolerances were chosen for this project to make the parts easy to manufacture while still fitting together accurately. A general range of ± 0.0005 – 0.05 (mainly ± 0.005 – 0.05) inches was used so that the compartments, lid, and hinge could align properly without being too tight or loose. These tolerances work well for materials like acrylic that are cut or machined using common fabrication tools such as laser cutters. They are precise enough to keep the assembly clean and functional but not unnecessarily strict for a simple, low-risk medical device. Acrylic was selected because it is lightweight, transparent, and durable, which makes it ideal for a meds box that needs to be both sturdy and easy to use. The clear surface allows users to see their medication without opening the lid, and the material is also resistant to moisture and easy to clean. Overall, the combination of these tolerances and the acrylic material help the device stay reliable, affordable, and safe for everyday use.

Specifically for the hinge, this tolerance should be described as follows. For the hinge hole I created on the hinge bases on the lid and main body, I used a diameter of 0.063 inches (about 1.6 mm) and for the rod insert that the mate allows rotation for, I used a diameter of 0.055 inches (about 1.4 mm). The tolerance for the rod should ideally not be as large. Although as stated above the 0.005 tolerance would still work for this situation, testing the upper edge (with rod diameter being 0.060 inches which is still less than the hole) could lead to possible real-world issues as the gap for rotational motion could be too small for ideal motion. Therefore, I would recommend keeping the tolerance for the hinge rod complex between -0.0025 and $+0.0025$.

Figure 11: Main Lid and Hinge Part Drawing

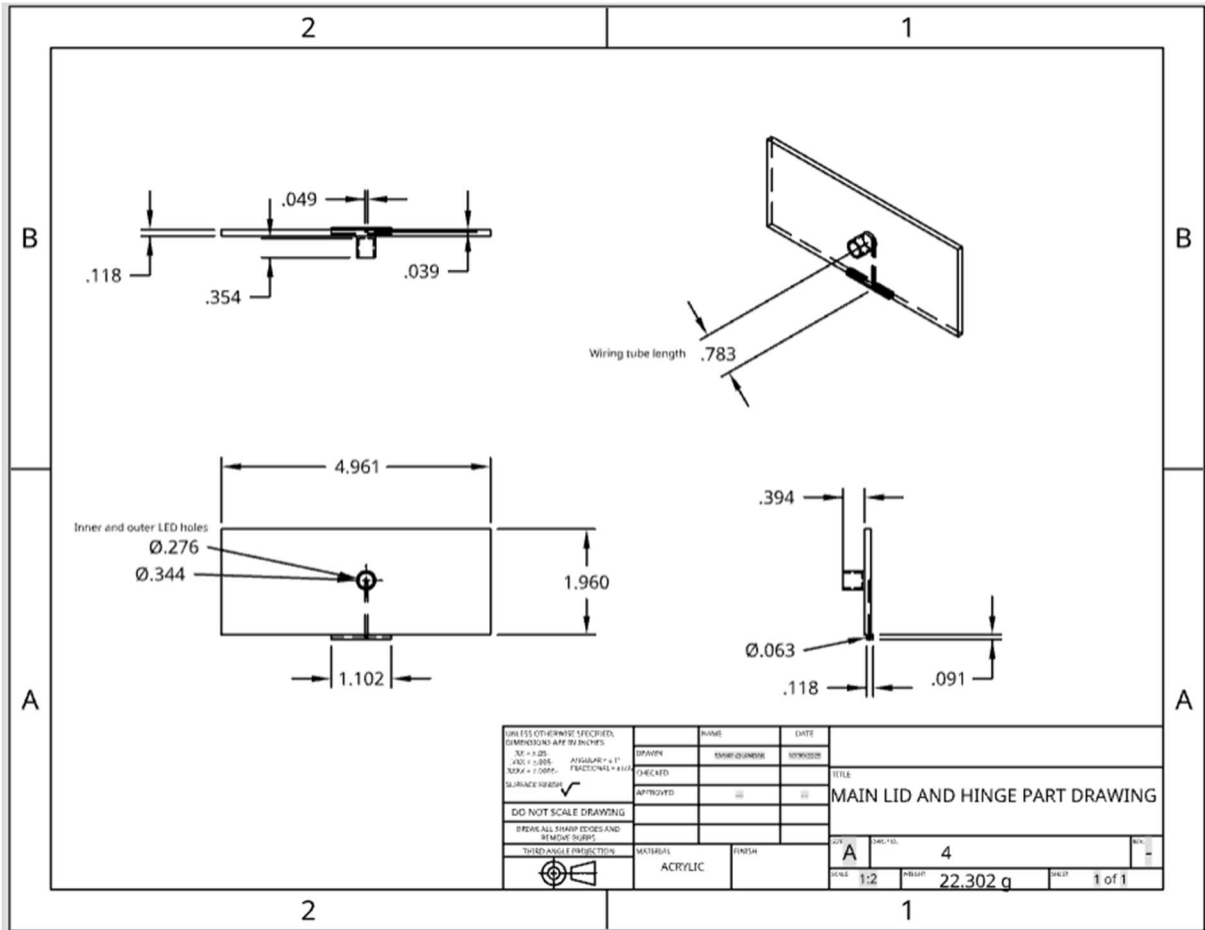


Figure 11 is a technical drawing presents the detailed dimensions and configuration of the main lid and hinge assembly used in the smart medicine container. The specified projections illustrate hinge hole placements, wall thicknesses, and the overall length and width of the lid to ensure precise alignment with the primary body. The isometric view highlights the hinge attachment point and wiring hole integration, allowing for the connection of the LED and Arduino components. The part is specified in acrylic to provide durability, light weight, and visibility of the internal compartments. Again, the tolerances are constant throughout given specific decimal points of each dimension.

Figure 12: Hinge Rod Part Drawing

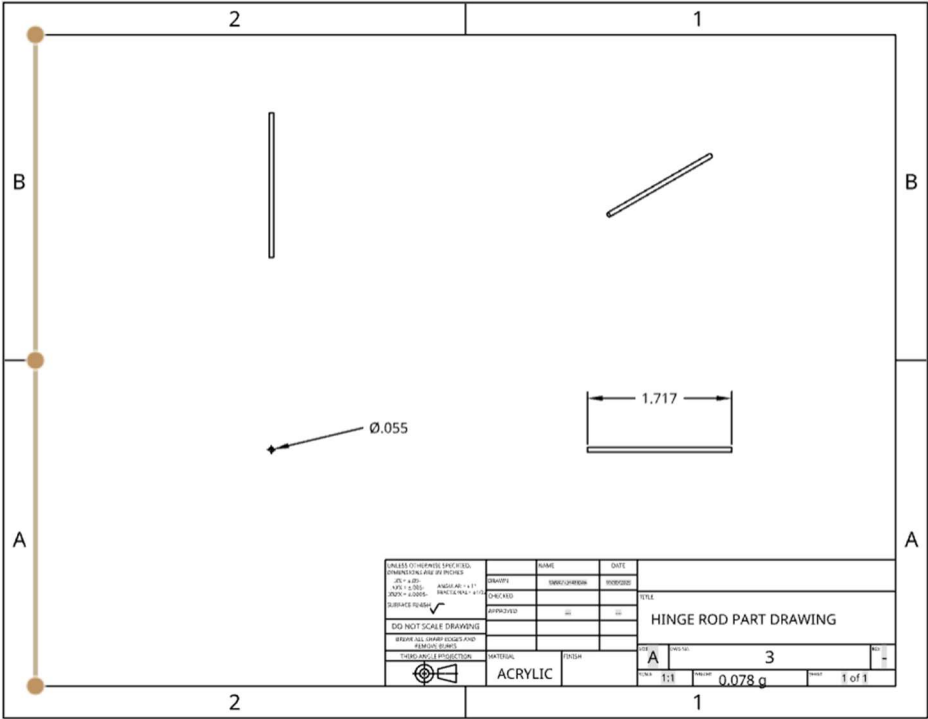


Figure 12 displays the required dimensions for the hinge such that it adequately fits inside the hinge bases on the lid and main compartment with room for tolerance.

Figure 13: Arduino Lid Part Drawing

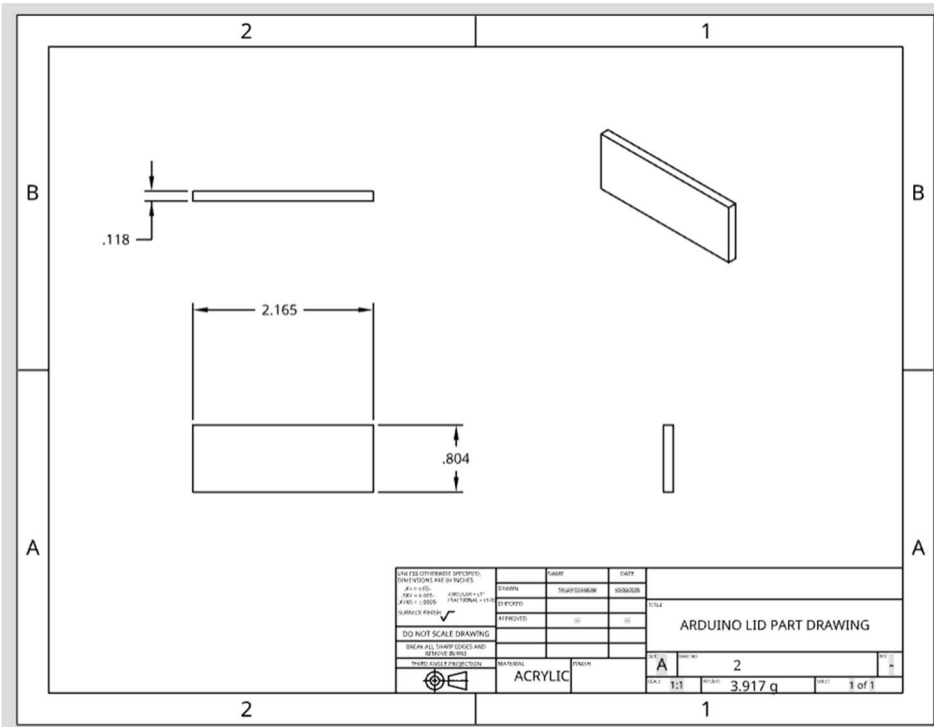
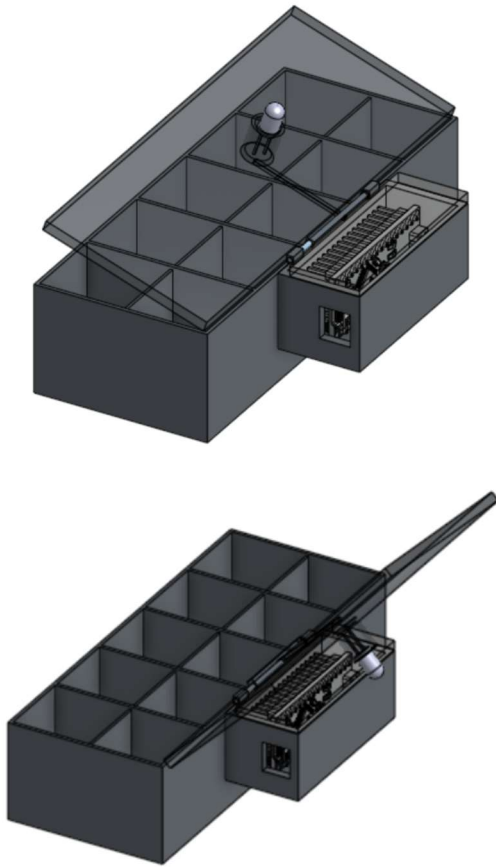


Figure 14: Display of Motion and Assembly



Assembly:

As described previously, the revolute and cylindrical mates are used to create the rotational movement displayed above in figure 14. The combination of these mates allows for the rotational movement of the lid between closing to about 150 degrees of openness. I did not use any mirrored parts and each individual component was created separately.

Discussion:

A couple strengths of my design are:

- **Modular and Accessible Layout:** The hinged transparent lid allows for easy visibility of the internal components and maintains safety and protection of internal components. 120 degrees opening and closing enables easy access for individuals with hand mobility issues as well.

- Integrated Electronics housing: The Arduino Nano system is integrated and built into the medical device with its own compartment and opening to allow easy access to USB or other input information. Provides safety from electronic malfunctions as well.
- Automated Medication Reminders: The device's main purpose and goal is that it identifies (through the Arduino code necessary) ideal or specific times the user should take their medicine and a flashing red LED sitting on top of the container lid will begin flashing indicating to the user that it is time to take their medicine.
- Easy to design, alter, and print: The design is purposefully created in a more rectangular or simplistic design with no soft curves (unless necessary) so that users who would like to alter the design can do so relatively easily. In addition, acrylic, the material, is relatively cheaper and can be easily used for manufacturing purposes. With the built-in complex Arduino system and wiring capabilities, additional sensors or electronic elements can be implemented to further improve the design.

A couple weaknesses of my design:

- The device is only divided into 4 physical parts in the CAD software (Onshape) which would make large scale alterations to major parts of the device somewhat difficult unless the user takes apart the whole design and re-makes it. This limited part assembly also can make the drawings and dimensional analysis of the device parts more difficult to visualize and understand for some individuals.
- Not very portable: The device is built to be more of a stationary instrument rather than portable with this design as the Arduino and LED system could be obstructive and inefficient to carry around.
- Hinge resistance: The hinge can be introduced to significant resistance and wear upon repeated use since the acrylic material will face large friction.
- Arduino Code required: The current design required Arduino code to be input into the nano system on the side of the box. Without proper coding input, the device will not function as designed to do so. The Arduino and LED are also not 3-D printable (normally) so those will need to be purchased separately.
- No feedback system: Unless further edits and changes are made to the design, the current model has no way of detecting whether the user has taken the medicine or not yet and thus will not know when to stop flashing. A simple fix for this issue could be the incorporation of an accelerometer which can detect movement of the lid in the rotational manner.

Additional components required: Arduino Nano, Wiring, Red LED, Accelerometer (optional), Arduino coding, and possibly incorporation of a locking mechanism to prevent the medicine box from opening on its own.

To enhance reliability and accessibility, several additional components should be incorporated into the Arduino-based medicine box. A buzzer would introduce an auditory reminder, improving

accessibility. A push button would provide a means to acknowledge medication intake, closing the feedback loop. Integration of a real-time clock module ensures precise reminder scheduling, while a battery indicator and power switch enhance safety and power management. Furthermore, clear labeling or a small display would improve usability and reduce confusion. These additions collectively make the device more robust, inclusive, and user-centered, aligning it more closely with real-world medical device standards.

Arduino NANO File Link: <https://grabcad.com/library/arduino-nano--1>

Red LED File Link: <https://grabcad.com/library/5mm-rgb-diode-led-1>